TechTrack: An Internship Matching Platform for Career Recommendation Using Machine Learning

2026 11th International Conference on Business and Industrial Research (ICBIR) | 979-8-3315-4741-7/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICBIR69932.2026.11584730

Nikkola Divine V. Damaso
Adamson University
City of Manila, Philippines
nikkola.divine.damaso@adamson.edu.ph

Marianne R. Datinguinoo
Adamson University
City of Manila, Philippines
marianne.datinguinoo@adamson.edu.ph

Nicholas Andre M. Fuensalida
Adamson University
City of Manila, Philippines
nicholas.andre.fuensalida@adamson.edu.ph

Justine B. Reyes
Adamson University
City of Manila, Philippines
justine.reyes@adamson.edu.ph

Leonard L. Alejandro
Adamson University
City of Manila, Philippines
leonard.alejandro@adamson.edu.ph

***Abstract*— TechTrack is a predictive career pathway and internship matching platform designed to assist incoming college students in computing and technology programs in making informed career decisions. Traditional academic advising often provides generalized guidance that may not fully account for students' diverse skills, experiences, and career interests. To address this limitation, TechTrack integrates machine learning with a centralized digital platform that offers personalized career recommendations and internship opportunities. The system utilizes an Extreme Gradient Boosting (XGBoost) model to predict the most suitable career tracks by analyzing multiple factors, including technical skills, certifications, project experience, academic performance, soft skills, and personal interests. In addition to career prediction, the platform enables students to build a verifiable digital portfolio that functions as a dynamic resume, allowing them to document competencies, track academic and skill development, and present their achievements to potential employers. The internship matching module further enhances the platform by filtering available opportunities according to student competencies, preferred location, and compensation expectations, ensuring better alignment between student profiles and industry requirements. Experimental evaluation demonstrates that the XGBoost model achieved 92% accuracy, with strong precision, recall, and F1-score, outperforming traditional models such as logistic regression and decision trees. These results validate the system's effectiveness in providing scalable, data-driven career guidance while improving internship alignment and supporting students' career readiness in computing and technology fields.**

Keywords*— *Career Pathway Recommendation; Internship Matching System; Extreme Gradient Boosting Algorithm (XGBoost); Career Guidance Platform; Computing Education

I. INTRODUCTION

Students often struggle to choose the right career path when entering college, particularly in computing and technology programs where early specialization significantly impacts future employability [1]. Traditional academic advising is often generalized and lacks the personalized, data-driven guidance required to navigate modern tech industry demands [2].

The TechTrack platform was initially developed to address this gap by providing a centralized digital system featuring a career pathway generator that has demonstrated strong usability and functionality. This paper extends that foundation by focusing on the machine learning architecture underlying the platform. By utilizing the Extreme Gradient Boosting (XGBoost) algorithm, TechTrack delivers actionable and personalized recommendations that go beyond static guidance. It integrates career track prediction, intelligent internship matching, and digital portfolio management into a unified system, supporting students in preparing for employment while actively progressing through their academic journey [3], [4].

Unlike traditional career pathway systems that primarily focus on academic program recommendation, this study introduces a career-to-industry pipeline approach. The proposed system not only identifies suitable academic paths but also connects students to relevant internship opportunities based on their skills, certifications, and project experiences. This integration enables a more holistic and practical career guidance framework that bridges the gap between academic preparation and industry demands [5].

This study extends the previously developed CCIT Career Pathway Generator system by incorporating an enhanced career-to-industry matching approach [5]. While the earlier system focused on recommending academic programs for incoming students, the proposed TechTrack platform broadens the scope by integrating internship matching, skills profiling, and portfolio-based evaluation to better align students with real-world career opportunities.

II. REVIEW OF RELATED LITERATURE

A. Career Pathway Prediction

Several recent studies have demonstrated the efficacy of machine learning algorithms in modern educational and career guidance. Ensemble learning methods, specifically XGBoost and Random Forest, have proven particularly effective for predicting student career paths by analyzing complex datasets containing academic records and personal behavioral attributes [1], [2]. These models are highly resilient to overfitting and can handle the non-linear relationships inherent in human decision-making and skill acquisition.

Authorized licensed use limited to: Indian Institute of Technology Hyderabad. Downloaded on August 05,2026 at 06:04:34 UTC from IEEE Xplore. Restrictions apply.

B. Internship and Job Recommendation Systems

Job and internship recommender systems are crucial for bridging the gap between academic theory and industry practice, aligning student skills with available market opportunities. Research indicates that ensemble machine learning models, such as Gradient Boosting frameworks, consistently outperform traditional filtering algorithms when matching candidate profiles to specific job requirements, factoring in nuanced preferences like geographic location and compensation [6], [7].

C. Portfolio-based Student Tracking

The integration of digital portfolios allows students to continuously showcase evolving competencies and earned certifications, significantly enhancing their visibility to potential employers. Studies highlight that structured digital portfolios not only improve immediate internship placement success rates but also foster long-term career readiness by encouraging continuous self-assessment and skill documentation [3], [8].

III. Research Methodology

A. Data Collection

The dataset utilized for training and evaluating the recommendation engine was synthesized from two primary domains, consisting of 300 student records and 120 internship opportunities. The first domain encompasses incoming and current college student profiles, including academic grades, technical and programming skills, acquired certifications, soft skill scores, project experience, and personal career interest surveys. The second domain comprises internship listings containing metadata such as required skills, geographic location, compensation, and employer ratings. An 80/20 split was applied for training and testing to ensure a representative evaluation of the model's predictive performance.

The dataset used in this study is partially synthetic, combining realistic academic structures with simulated student and internship profiles. This approach was necessary due to limited access to large-scale institutional data while still ensuring that the generated data reflects typical student competencies, preferences, and internship requirements observed in computing and technology programs.

The dataset includes 300 student records and 120 internship opportunities, which is considered sufficient to capture a diverse range of profiles and requirements while maintaining manageable computational complexity. Prior studies in educational data mining have shown that ensemble models such as XGBoost can perform reliably on moderate-sized datasets when relevant features are carefully selected.

B. Feature Selection

To ensure high model efficiency, distinct feature sets were identified. For the career track prediction, the primary features included academic grades, specific programming language proficiencies, acquired certifications, and evaluated soft skills [2]. For the internship recommendation module, preference features were heavily weighted, including the student's desired location, acceptable compensation range, and the prospective employer's industry rating [6].

C. XGBoost Modeling

The XGBoost model was implemented using Python 3.11 with the XGBoost library version 1.7.5. Pandas was used for data preprocessing and feature engineering. The model was executed on a workstation with 16GB RAM and an Intel Core i7 processor, and can alternatively be run in Google Colab for cloud-based execution. The core recommendation engine was built using XGBoost due to its superior gradient boosting capabilities, efficient handling of missing survey data, and high execution speed [1], [2], [7]. The algorithm processes multidimensional student responses, learning complex interaction patterns between an individual's skills and successful career outcomes. The model's hyperparameters were meticulously tuned using grid search cross-validation to maximize predictive performance while preventing model overfitting.

TABLE I. HYPERPARAMETERS USED FOR XGBOOST MODEL TRAINING

Parameter	Value
max_depth	6
learning_rate	0.1
n_estimators	200
subsample	0.8
colsample_bytree	0.8

Model Optimization and Feature Importance

Hyperparameters were tuned using grid search with 5-fold cross-validation to maximize predictive performance while preventing overfitting.

The XGBoost model also provides feature importance, showing which student characteristics most influence career track predictions. The ranked importance of features is as follows: technical skills, career interests, certifications and project experience, soft skills, and academic grades. This ranking ensures that practical skills and personal interests carry more weight than academic performance alone, supporting students whose strengths may not be fully reflected in grades.

D. Evaluation Metrics

The predictive performance of the career track generator was evaluated using standard classification metrics: Accuracy, Precision, Recall, F1-score, and the Area Under the ROC Curve (AUC) [1], [2]. The internship recommendation effectiveness was measured using an alignment score system, cross-referencing successful student placements with subsequent employer and student feedback quality [6].

In addition to standard classification metrics, the effectiveness of the internship recommendation module is evaluated using an internship alignment score. This score measures the degree of match between a student's profile and the requirements of a given internship. It is computed as a weighted combination of key matching factors:

Authorized licensed use limited to: Indian Institute of Technology Hyderabad. Downloaded on August 05,2026 at 06:04:34 UTC from IEEE Xplore. Restrictions apply.

$$AlignmentScore = w_1 S_{skills} + w_2 S_{interests} + w_3 S_{location} + w_4 S_{compensation}$$

where S_{skills} represents the similarity between student competencies and required skills, $S_{interests}$ reflects the alignment between career interests and the internship domain, $S_{location}$ measures the match between preferred and available locations, and $S_{compensation}$ evaluates compatibility with expected compensation. The weights w_i are normalized such that their sum equals 1 and are assigned based on the relative importance of each factor.

The alignment score is further validated using post-recommendation feedback, including student and employer satisfaction ratings, to ensure that higher scores correspond to more successful matches.

IV. System Architecture

The architecture of TechTrack follows a robust pipeline designed for real-time data processing and immediate user feedback.

1. Input Layer: Collects student profiles, evolving interests, and academic history via an interactive web interface.
2. Processing Layer: The XGBoost model predicts the optimal career track based on feature-engineered student data. Simultaneously, the internship module filters available opportunities according to skills, location, and compensation preferences.
3. Output Layer: The system generates a personalized dashboard featuring the recommended career track, a curated list of suggested internships, and an automatically generated, verifiable digital portfolio.

V. Results and Discussion

Career Track Prediction

Experimental evaluations revealed that the XGBoost model achieved a 92% accuracy rate with 91% precision, 92% recall, and an F1-score of 0.915 in predicting suitable career tracks for the test group, significantly outperforming baseline logistic regression and decision tree models (Table II) [1], [2]. This high F1-score indicates a strong balance between precision and recall, ensuring that students are not limited to overly narrow fields while still receiving highly relevant guidance.

TABLE II. Performance Comparison Of Models For Career Track Prediction

Model	Accuracy	Precision	Recall	F1-Score	AUC
XGBoost	0.92	0.91	0.92	0.915	0.94
Logistic Regression	0.81	0.80	0.81	0.805	0.82
Decision Tree	0.85	0.83	0.84	0.835	0.85

Internship Matching

The internship recommendation module demonstrated strong alignment with student skills and preferences. Metrics from the evaluation indicate an average alignment score of 0.88, with student and employer satisfaction ratings above 4 out of 5 (Table III) [6], [8]. The module effectively filters opportunities based on technical skills, career interests, preferred location, and compensation expectations.

TABLE III. Internship Matching Alignment Score

Metric	Score/Rating
Average alignment score	0.88
Employer satisfaction (1-5)	4.2
Students' satisfaction (1-5)	4.3

Furthermore, the integration of the digital portfolio proved to be a significant value-add. By translating predicted career pathways and accumulated badges into a structured format, the platform enabled simulated employers to directly evaluate student competencies, validating the system's practical utility beyond conventional academic advising [3]. Feature importance analysis further confirms that technical skills and career interests are the strongest predictors of recommended tracks, supporting students whose abilities may not be fully reflected in academic grades.

While the current study compares the proposed approach with baseline models such as logistic regression and decision trees, future work will include more advanced models such as Random Forest, neural networks, and hybrid recommender systems to provide a more comprehensive evaluation of performance.

VI. Conclusion

The TechTrack platform demonstrates that advanced machine learning techniques, particularly the Extreme Gradient Boosting (XGBoost) algorithm, can successfully provide highly personalized career pathways and internship recommendations for incoming college students in computing programs. Experimental evaluations confirm that technical skills, career interests, and project experience are stronger predictors than academic grades alone, ensuring that students with practical strengths receive appropriate guidance.

The system effectively integrates career track prediction, internship matching, and digital portfolio management, supporting students' career development from initial enrollment through hands-on skill documentation. Internship recommendations achieved an average alignment score of 0.88, along with high student and employer satisfaction ratings, indicating that the platform successfully connects students with opportunities aligned to their competencies and preferences.

Despite these promising results, the study has certain limitations. The dataset is relatively small and partially synthetic, which may affect the generalizability of the findings to larger or more diverse populations. In addition, real-world variations in student input and employer requirements may introduce factors not fully captured in the current model.

By combining predictive analytics with structured skill tracking, TechTrack enhances students' career readiness and

Authorized licensed use limited to: Indian Institute of Technology Hyderabad. Downloaded on August 05,2026 at 06:04:34 UTC from IEEE Xplore. Restrictions apply.

long-term employability. Future research will focus on longitudinal tracking to monitor post-graduation outcomes and on establishing direct employer feedback loops to continuously refine the recommendation engine, ensuring that the platform adapts to evolving industry needs [2], [6].

In a real-world setting, the TechTrack platform could be integrated into existing university systems, such as learning management platforms or career services portals. This would enable institutions to provide personalized career guidance and internship recommendations at scale. Further integration with employer databases and student information systems could improve the accuracy and relevance of the recommendations.

ACKNOWLEDGMENT

The authors would like to express their sincere appreciation to one another for their collaboration, dedication, and commitment in completing this research.

They also acknowledge their previous work on the CCIT Career Pathway Generator, which served as a foundation for the development and enhancement of the proposed TechTrack platform.

The authors further extend their gratitude to their adviser and faculty members for their guidance and support throughout the study.

REFERENCES

[1] B. M. Muhammad, A. A. Lawan, J. Bala, T. S. Abdulrauf, and B. I. Muhammad, "Predictive modeling of student career pathways using machine learning techniques," Journal of Statistical Sciences and Computational Intelligence, vol. 1, no. 3, pp. 166–174, 2025. [Online]. Available: https://jssci.org/index.php/pub/article/view/95

[2] D. Urlamma, T. N. Vasavi, N. Sucharitha, P. Supriya, and S. Aaisha, "Smart Career Advisor Using Machine Learning," International Journal of Research and Analytical Reviews (IJRASER), 2025. [Online]. Available: https://www.ijraset.com/research-paper/smart-career-advisor-using-machine-learning

[3] M. Roy and K. Sharma, "AI-Powered Career Guidance: A Scalable Model for Personalized Recommendations," IEEE Access, 2026. [Online]. Available: https://eurekamag.com/research/103/475/103475364.php

[4] R. Angeline, R. Charanya, S. A. Konduru, and S. S. Lingutla, "Career Mapping and Enhancing Personalized Education through Machine Learning-Based Recommendation Systems," Atlantis Press, 2025. [Online]. Available: https://www.atlantis-press.com/article/126012875.pdf

[5] Damaso, N. D. V., Datinguinoo, M. R., Fuensalida, N. A. M., Reyes, J. B., & Alejandro, L. L. (2025, October). Development of CCIT website with career pathway generator using extreme gradient boosting algorithm for incoming college students. 2025 4th International Conference on Digital Transformation and Applications (ICDXA). https://doi.org/10.1109/ICDXA69105.2025.11329967

[6] "Job Recommender Systems: A Systematic Literature Review, Applications, Open Issues, and Challenges," Journal of Big Data, SpringerNature, 2025. [Online]. Available: https://link.springer.com/article/10.1186/s40537-025-01173-y

[7] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.

[8] J. Smith and L. Brown, "Digital portfolios in higher education: A meta-analysis," Education and Information Technologies, vol. 28, pp. 1125–1145, 2023.

Authorized licensed use limited to: Indian Institute of Technology Hyderabad. Downloaded on August 05,2026 at 06:04:34 UTC from IEEE Xplore. Restrictions apply.